

IMPORT DATA USING TRANSFORM MAPS (SPREADSHEET)

Project Type: Micro Project

Platform: ServiceNow

Data Source: Excel / Spreadsheet

Main Concepts: Import Sets, Transform Maps, Coalesce, Reports and Dashboards

Description: This micro project focuses on importing structured data from an external spreadsheet into the ServiceNow platform using Import Sets and Transform Maps. The project simulates a real-world scenario where bulk employee data is received in Excel format and must be migrated into ServiceNow efficiently and accurately. The spreadsheet data is first loaded into an Import Set table, which serves as a staging area, and a Transform Map is then configured to map source fields to the appropriate target table fields. Upon executing the transformation, records are created in the target table and validated to ensure data integrity. This project helps beginners understand the end-to-end data import process in ServiceNow, including data loading, field mapping, transformation execution, and result verification.

1. PHASE 1: PREPARE THE SPREADSHEET

- Open a new Google Spreadsheet.
- Create the spreadsheet with sample employee data.
- Download the spreadsheet in Sample Spreadsheet.xlsx format.

2. PHASE 2: CREATE A NEW CUSTOM TABLE

Navigation: Tables → Create New

- Label: Employee Test
- Name: u_employee_test
- After creating and saving the Employee Test table, open the table form.
- Use Form Layout to create the required fields.
- Create Employee ID – Type: String.
- Create Employee Name – Type: String.
- Create Email – Type: String.
- Create Department – Type: String.
- Create Location – Type: String.
- Save the form and verify that all fields are displayed.

3. PHASE 3: CREATE THE IMPORT SET TABLE

Navigation: Application Navigator → Load Data

- Click Load Data and create an import table.
- Label: Employee Import.
- Name: u_employee_import (auto populated).

- Click Submit.
- After submission, create the Transform Map.

4. PHASE 4: CREATE THE TRANSFORM MAP

- Create the Transform Map from the Load Data process.
- Name: Sample Spreadsheet Import.
- Source Table: Employee Import.
- Target Table: Employee Test.
- Click Auto Map Matching Fields.
- Use Mapping Assist to map the source fields to the target fields.
- Verify that the required source and target fields are mapped.
- Click Save.
- Click Transform and execute the transformation.
- Verify that the transformation completes successfully.

5. PHASE 5: TRANSFORM DATA AND VALIDATE

- Open the Employee Test table from the Application Navigator.
- Verify that the imported employee records are created in the Employee Test table.
- Use Personalize List Columns to arrange the fields if required.
- The imported records should be available in the target table.
- Final result: employee records are successfully imported into ServiceNow using Import Sets and Transform Maps.

6. PHASE 6: ENABLE COALESCE TO AVOID DUPLICATE RECORDS

Coalesce is used when the same employee data may be imported multiple times. It helps the system identify existing records and update them instead of creating duplicate records.

- Open the created Transform Map: Sample Spreadsheet Import.
- Open the Field Maps section under the related list.
- Select a suitable unique field, such as Employee ID.
- Set Coalesce to True for that field.
- Save the Transform Map.
- Import another Excel file containing existing and new employee data.

7. PHASE 7: INSERT NEW DATA IN EXCEL FORMAT

- Navigate to All menus and search for Load Data under System Import Set.
- Open Load Data.
- Select the existing Employee Import table.

- Select File as the source of import and choose the Excel file.
- Sheet Number: 1.
- Header Row: 1.
- Click Submit.
- Run the Transform using the previously created Transform Map.
- Verify that the state is Completed and Successful.
- Open Transform History to review inserted, updated, skipped or ignored records.

In the project example, four rows were uploaded. Two new employee IDs were inserted and two existing employee IDs were updated with changed employee names and email addresses. When the same sheet was uploaded again without changes, the records were ignored because the existing data had already been processed.

8. PHASE 8: CREATE REPORTS ON EMPLOYEE TEST TABLE

Report 1: Employees by Department

- Report Name: Employees by Department.
- Source Type: Table.
- Table: Employee Test.
- Chart Type: Pie Chart.
- Group By: Department.
- Aggregation: Count.
- Run and save the report.

Report 2: Employees by Location

- Report Name: Employees by Location.
- Table: Employee Test.
- Chart Type: Bar Chart.
- Aggregation: Count.
- Group By: Location.
- Run and save the report.

Report 3: Employee List Report

- Report Name: Employee List Report.
- Table: Employee Test.
- Type: List.
- Columns: Employee ID, Employee Name, Email, Department and Location.
- Run and save the report.

9. PHASE 9: ADD REPORTS TO DASHBOARD

- Open All menus and search for Dashboards.

- Open Dashboards and create a new dashboard.
- Dashboard Name: Employee Analytics Dashboards.
- Open the created reports.
- Use the Share option and select Add to Dashboards.
- Select Employee Analytics Dashboards and click Add.
- Add Employees by Department, Employees by Location and Employee List Report to the dashboard.
- Reports can also be added as widgets using the plus (+) option.
- Users, groups and roles can be restricted through the Share option.

10. PROJECT OUTCOME

The project implements an employee data management process using ServiceNow Import Sets and Transform Maps. Spreadsheet data is loaded into a staging Import Set table, mapped to the Employee Test target table and transformed into ServiceNow records.

Coalesce provides duplicate prevention by identifying existing employee records using a unique field such as Employee ID. Existing records can therefore be updated rather than duplicated when the same employee data is imported again.

Reports and dashboards provide visibility into employee information and import activities. The project includes department-wise employee distribution, location-wise employee distribution, employee list information, newly added employees, updated records and import results.

11. CONCLUSION

This project successfully implemented an automated employee data management solution using ServiceNow Import Sets and Transform Maps, ensuring accuracy, efficiency and data integrity throughout the employee onboarding and update process.

The use of Coalesce is a key enhancement because it prevents duplicate records and allows repeated imports to update existing employee data. This keeps the employee database clean, reliable and consistent.

The addition of Reports and Dashboards provides centralized visibility into employee data, import performance and data quality. The dashboard allows administrators and HR teams to monitor employee information through reports and visual representations, reducing the need for manual analysis.